

Subsurface imaging in shield tunnelling and pipeline localisation via a robust cosmic-ray muography system: a field study

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ABSTRACT: Unforeseen geological conditions and buried utilities pose major risks to urban shield tunnelling, whereas borehole and surface investigations provide only limited coverage in dense urban areas. Cosmic-ray muography images subsurface density passively, but its use on construction sites requires instruments that withstand vibration, heat, humidity and restricted access. This paper presents a field-deployable three-layer Micromegas muography system and two applications on Shenzhen Metro Line 22. The system combines environmental control, pneumatic vibration isolation, a low-consumption gas supply, remote condition monitoring and dedicated data acquisition, and supports both TBM-mounted and stationary measurements. Mounted on two tunnel boring machines (TBMs) for 1371 rings, it operated unattended with 97% data-acquisition availability once external power and network outages were excluded; TBM advance increased the raw data rate, but the reconstructed muon rate remained unchanged. In the first application, muon angular distributions measured at two rings were combined with borehole logs to extend the borehole-derived density model into three dimensions, indicating a possible low-density zone above and to one side of the tunnel. In the second, an abandoned pipeline that ground-penetrating radar had failed to locate was surveyed from four stations in the completed parallel tunnel, and its axis was estimated with station-specific U-Net models and multi-view geometric reconstruction. The vertical alignment of the tunnel was modified on this basis and the shield passed the crossing without interference, although the pipeline position was not verified directly. The results show that muography is feasible on construction sites and can support precautionary decisions as a targeted screening tool; shorter exposures, matched reference measurements and uncertainty quantification are needed for routine use.

Keywords: muography; shield tunnelling; density reconstruction; pipeline localisation

1 INTRODUCTION

Unforeseen geological conditions and buried utilities are among the principal risks in urban shield tunnelling: weak zones, cavities or obstructions ahead of the tunnel face can interrupt advance and damage adjacent infrastructure. Borehole investigations provide only discrete point data, and surface geophysical surveys are often restricted in dense urban areas. Cosmic-ray muography offers a complementary means of subsurface imaging from within the tunnel. Atmospheric muons penetrate tens to hundreds of metres of soil and rock, and an underground detector measures the angular distribution of the transmitted muon flux; comparison with an open-sky reference yields the opacity, or density integrated along each muon trajectory (Procureur 2018). Combining observations from several viewing directions and detector positions with the tunnel geometry and borehole data enables three-dimensional imaging of density variations around and ahead of the excavation. The technique is passive and non-invasive: it requires no artificial radiation source, surface access or contact with the target, and can operate concurrently with construction.

Previous underground studies have reconstructed overburden density and rock–air interfaces from multiple detector locations (Guardincerri et al. 2017), identified concealed construction

shafts and a subsequently confirmed overburden void in a railway tunnel (Thompson et al. 2020), and detected structural changes along a subway station and tunnel (Mao et al. 2023). These studies demonstrate the potential of in-tunnel measurements, but sustained deployment during active shield construction imposes additional instrumentation and operational requirements.

A detector mounted on an operating tunnel boring machine (TBM) is exposed to vibration, heat and humidity in a confined space with limited maintenance access, and its observation geometry changes continuously as the shield advances. Stationary investigations, by contrast, require accurate repositioning of the instrument so that observations of a fixed target from several stations can be combined.

This paper presents a field-deployable three-layer Micromegas muography system that supports both dynamic TBM-mounted measurements and stationary multi-station investigations. The contributions are: (1) an instrument designed for operation under construction conditions; (2) an assessment of its long-term operation on Shenzhen Metro Line 22; and (3) two field demonstrations, namely TBM-mounted ground-density imaging and a four-station investigation of an unlocated buried pipeline.

2 TUNNEL-DEPLOYABLE MUOGRAPHY SYSTEM

2.1 Detector and measurement principle

The tracking unit comprises three thermally bonded Micromegas layers, each with a $40 \times 40 \text{ cm}^2$ sensitive area and separated by approximately 15 cm (Feng et al. 2021). Orthogonal readout strips segmented at 1.6 mm give two-dimensional hit positions on each layer, and each muon trajectory is reconstructed from the hits in all three layers. This fine segmentation provides an angular resolution of about 0.3° , among the best available for a detector with such a large sensitive area, so that fine angular structure is resolved without sacrificing geometric acceptance. The front-end and data acquisition electronics were developed in-house and customised for muography, forming inter-layer coincidences and aggregating events in a compact readout chain (Wang et al. 2025; Liu et al. 2025). Table 1 summarises the main specifications of the deployed system.

Table 1. Main specifications of the muography system.

Item	Value	Item	Value
Detector	Micromegas, 3 layers	Main enclosure	$700 \times 450 \times 800 \text{ mm}$, 85 kg
Sensitive area	$400 \times 400 \text{ mm}$ per layer	Gas enclosure	$350 \times 330 \times 750 \text{ mm}$, 25 kg
Layer spacing	150 mm (300 mm baseline)	Protection	IP56
Readout	X–Y strips, 1.6 mm pitch	Power	220 V AC, 800 W
Layer efficiency	> 95% (nominal)	Data interfaces	Ethernet, RS485
Angular resolution	0.3° (nominal)	Inclination	35° , adjustable
Max. event rate	3 kHz	Monitoring	Temperature, humidity, inclination, vibration

2.2 Design for the tunnel environment

Shield tunnelling exposes the instrument to heat and humidity, machine vibration, a continuously changing observation geometry and limited access for maintenance. The IP56 main enclosure (Figure 1a,b) integrates the detectors, readout electronics, power supplies and a sealed air-conditioning unit that controls internal temperature and humidity. For TBM-mounted operation, four pneumatic isolators (Figure 1c) separate the inclined detector frame from the shield support platform, and vibration sensors on the platform and inside the enclosure record the vibration on each side of the isolators. An inclinometer records the detector inclination so that each measurement can be associated with its observation geometry. A dedicated gas-control subsystem in a separate enclosure (Figure 1a) supplies the argon–carbon-dioxide mixture with much lower consumption than laboratory flow-gas operation; a single 8 L cylinder at 10 MPa sustained approximately three months of field operation, which set the routine maintenance interval. The system is an unattended

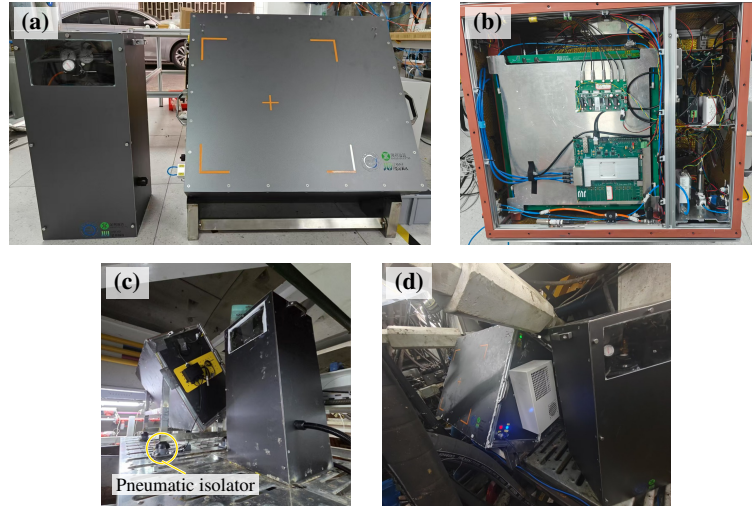


Figure 1. Muography system: (a) gas-supply (left) and main (right) enclosures; (b) main-enclosure interior; (c) right-line and (d) left-line TBM installations.

(headless) instrument: it runs without an operator on site, and its status, environmental and vibration data are monitored and controlled remotely over Ethernet or RS485, making it suitable for autonomous, long-term sensing in the tunnel.

2.3 Field deployment and operational performance

The field tests were carried out on a twin-tunnel running section of Shenzhen Metro Line 22 between November 2025 and July 2026, during which the system travelled with the shields through 1371 rings at depths of approximately 15–30 m. On the left line, a shorter campaign through deeper ground, it was installed about 5 m behind the cutterhead (Figure 1d); this campaign provided the vibration record below and Demonstration I. On the right line, a longer campaign at shallower depth, it was installed about 15 m behind the cutterhead (Figure 1c). Detector positions and orientations were linked to the ring records, and the enclosure remained below 40 °C and 70% relative humidity throughout.

The right-line campaign demonstrated stable long-term operation. Over nearly five months of unattended monitoring, data were acquired for 85% of the time, and for 97% once external power and network outages were excluded; the remainder was routine maintenance.

TBM operation disturbed the raw data stream but not the muon measurement (Figure 2). Advance and pause periods between May and July 2026 were identified from the platform vibration record. During advance, the platform acceleration rose sharply and the DAQ data rate rose with it, by 76% on average over 56 transitions: the isolators did not fully suppress the disturbance, and residual vibration, electromagnetic interference from the TBM, or both induced spurious triggers. The reconstructed three-layer muon rate, by contrast, changed by only -1.5% (95% interval -6.9 to $+5.8\%$), showing that the three-layer coincidence and track reconstruction rejected these triggers.

3 DEMONSTRATION I: DENSITY IMAGING AHEAD OF THE TBM

3.1 Engineering objective

Borehole logs describe the ground only at discrete points, and the strata between them are normally interpolated. The objective of this demonstration was to refine that interpolation with muon data and to extend the borehole-based density model into three dimensions around the shield. Because the inclined detector views the ground above and ahead of the machine, density anomalies can be indicated about 30–50 m in advance of the TBM, giving time for precautionary measures that reduce construction risk.

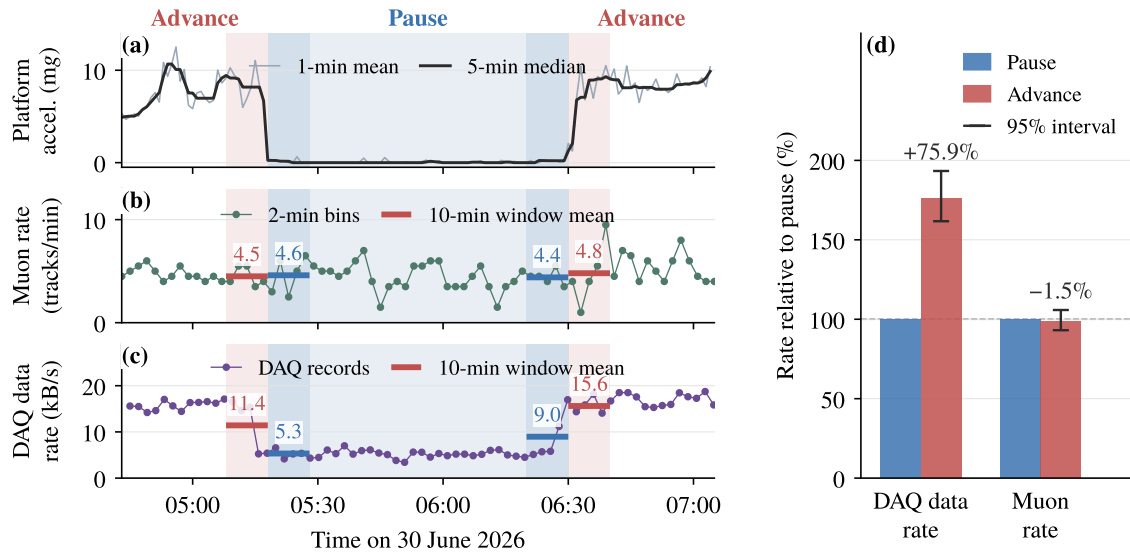


Figure 2. Detector response to TBM advance and pause. (a)–(c) A typical sequence: platform acceleration outside the isolators, reconstructed muon rate and DAQ data rate; bars are 10-min window means. (d) Pooled change during advance relative to the adjacent pause, with 95% intervals.

3.2 Test section

A test section near a road bridge, at a depth of approximately 25 m, was selected. Several borehole logs were available along this section, and the low advance rate allowed exposures of approximately 1–1.5 days at each of two rings, seven rings apart. These exposures are long relative to normal advance and therefore well suited to the demonstration. Between the two rings the detector moved approximately 13 m along the tunnel while its elevation and inclination changed only slightly, so the two measurements observe the same ground from different positions. Open-sky reference measurements were acquired with the same detector and track selection.

3.3 Density inversion method

The reconstruction starts from a prior density model built from the borehole logs. Layer boundaries are interpolated between boreholes along the tunnel and extended laterally, each layer is assigned a reference density, and the tunnel cavity and lining are modelled explicitly. For a trial density distribution, ray tracing calculates the path length through each material for every muon direction (Siddon 1985), and a muon spectrum (Guan et al. 2015) with standard energy-loss ranges (Groom et al. 2001) converts this into the expected muon count. The algorithm then adjusts three-dimensional density corrections, on 2 m voxels, until the expected counts best agree with the measured counts from both rings, accounting for their Poisson statistics.

Two viewing positions alone cannot constrain a three-dimensional density field, so the inversion relies on prior information. Corrections are penalised more strongly near boreholes, where the ground is known, and smoothing between neighbouring voxels is relaxed across interpreted layer boundaries, where real density contrasts are expected. A common normalisation, estimated from directions away from known bridge piles, accounts for differences between the underground and open-sky datasets. The result is therefore a regularised correction to the borehole model rather than an independent absolute-density image.

3.4 Results and interpretation

Figure 3 shows the joint reconstruction from both rings. The principal feature is a low-density zone above the tunnel and to one side, with a mean correction of -0.36 g cm^{-3} relative to the borehole model; a smaller positive correction appears near ring 1679. Because sparse viewpoints and regularisation can create artefacts, two checks test whether this feature is supported by the data. First, each ring was reconstructed on its own: both show a density deficit in the same region (-0.21 and -0.24 g cm^{-3}), so the feature does not depend on a single dataset, although its detailed

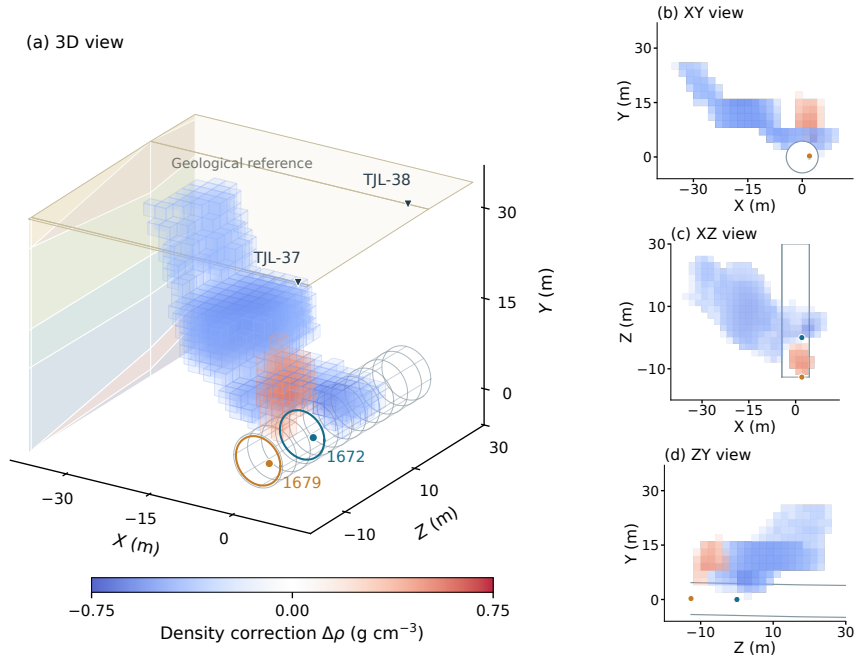


Figure 3. Joint density corrections relative to the borehole model: (a) three-dimensional rendering and (b–d) XY, XZ and ZY views. Only 2 m voxels with direct ray coverage and $|\Delta\rho| \geq 0.35 \text{ g cm}^{-3}$ are shown; transparency decreases with $|\Delta\rho|$.

shape differs. Second, each single-ring reconstruction was used to predict the counts recorded at the other ring. Corrections fitted only to statistical fluctuations would not improve this independent prediction; in both directions they agreed better with the measured counts than the borehole model alone, modestly but consistently.

The result therefore indicates a possible low-density zone rather than a confirmed void, and is best used to flag ground for attention ahead of the shield. Its confidence is limited mainly by the short exposure at each ring, the sparse borehole control and the non-simultaneous open-sky reference, which Section 5.2 returns to.

4 DEMONSTRATION II: DISUSED PIPELINE LOCALISATION

4.1 Engineering objective

An abandoned pipeline, buried approximately 17 m deep, was recorded on the site utility map but could not be found at the mapped position during site investigation. The right-line tunnel under construction was designed to pass only about 0.8 m beneath it, so its actual position was needed to judge the risk of interference. Ground-penetrating radar surveys failed to locate it correctly, and no other effective method was available; muography was therefore used to estimate its position.

4.2 Survey arrangement

The detector was deployed in the completed left-line tunnel, facing the right line under construction about 15 m away. It was placed in turn at four stations spanning approximately 11 m along the tunnel, each oriented towards the region where the pipeline was most likely to lie (Figure 4, top), with an exposure of approximately one day at each. A shared open-sky reference was used for all four stations, and the detector poses were surveyed for the reconstruction.

4.3 Pipeline reconstruction method

The workflow separated feature detection from three-dimensional localisation. First, a station-specific U-Net (Ronneberger et al. 2015) was trained on simulated muon angular distributions containing pipelines with varied locations and orientations, together with cases containing no

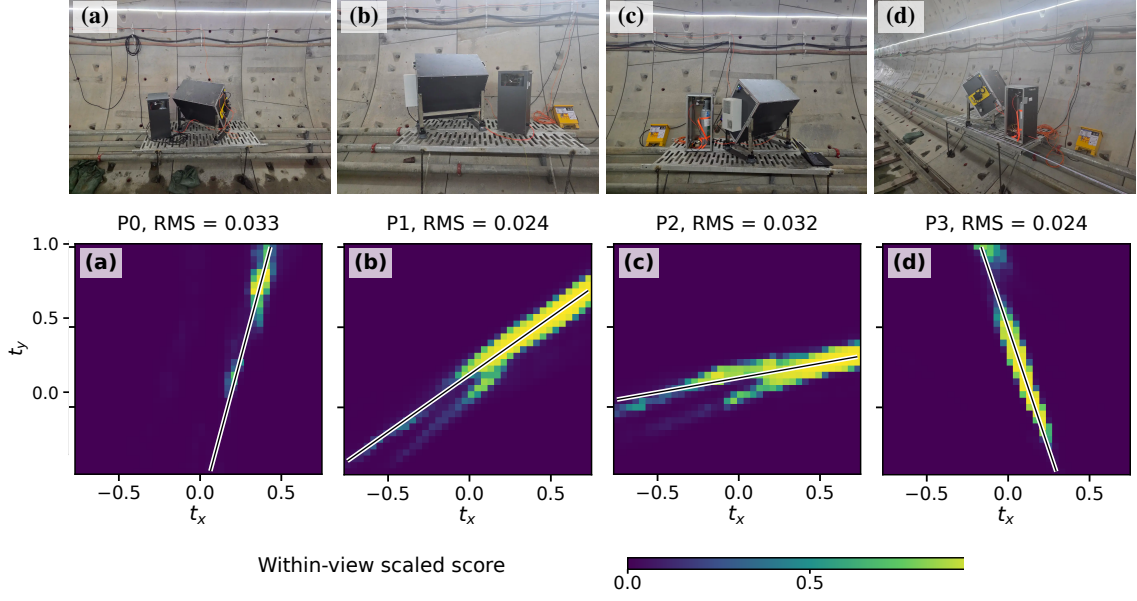


Figure 4. Pipeline survey at stations (a) P0 to (d) P3: detector deployment (top) and network score maps over the muon angular distribution, normalised per station, with fitted ridges (bottom). Axes are track slopes in detector coordinates; RMS is the residual about the ridge.

pipeline, and reached an area under the ROC curve above 0.95 on simulated test data. Each network converted the measured angular distribution into relative feature scores, which indicate pipeline-like structure but are not calibrated probabilities. All four measured score maps contained an extended ridge (Figure 4, bottom). A straight ridge was then fitted to the high-score region in each map, and the measured detector poses transformed the four ridges into backprojection planes in a common tunnel frame. The pipeline axis is the line most consistent with all four planes, with observations weighted by ridge quality and a soft preference for a near-horizontal direction reflecting the expected utility geometry.

4.4 Results and engineering use

The four observations converged on a single, nearly horizontal candidate axis crossing above the tunnel alignment (Figure 5), providing an estimate of the orientation and burial depth of the unlocated pipeline. On this basis, the vertical alignment of the right-line tunnel was modified, and the shield passed the crossing without interference. The modification was thus a precautionary decision based on an estimate: the pipeline position was not verified by boreholes or other direct means, and the clean passage is consistent with, but does not prove, the estimated axis.

5 DISCUSSION

5.1 Lessons for field instrumentation

The campaign shows that one transportable Micromegas system can support dynamic TBM-mounted and stationary multi-station measurements. TBM advance raised the raw data rate despite pneumatic isolation, but three-layer track reconstruction kept the muon rate unaffected; separating the vibration and electromagnetic contributions is a design task for future systems. In addition, the low-consumption gas supply allowed about three months between routine services. Most right-line downtime arose from external power and network interruptions, making site infrastructure part of the deployment design. A more integrated next-generation unit is under development, about 20% lighter and 10% smaller; an improved gas system allows a 1–2 L cylinder while keeping the same maintenance interval.

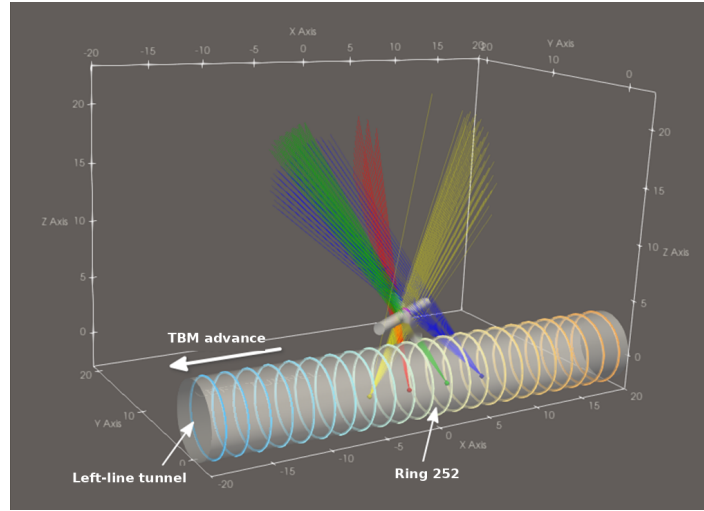


Figure 5. Three-dimensional pipeline reconstruction in the ring-252 frame (m). Coloured ray fans represent the backprojection planes from P0–P3, and the grey tube shows the candidate pipe with its recorded diameter of 0.8 m.

5.2 Integration into tunnelling practice

Figure 6 outlines how tunnel muography can be moved from field measurement to engineering decisions. Both applications would share the same measurement chain but use different interpretation branches, and each output should pass through an evidence review before informing a precautionary decision, with remaining uncertainty directing further measurement stations. In routine shield construction, an unattended detector on the TBM back-up gantry could screen the ground ahead of the face, with its data combined with borehole logs and TBM operating records, while stationary multi-station surveys from completed tunnels could address specific targets such as utilities, shafts or obstructions.

Excavation of one ring takes hours, depending on soil and ground conditions, whereas a statistically adequate angular distribution currently requires about one day of exposure at each station. Quasi-real-time, continuous warning is therefore challenging at this stage, and the system would be better suited to targeted screening of known concerns, such as suspected utility crossings or sections with sparse borehole control. Larger or multiple detectors, matched sky references, systematic pose records, automated reconstruction with uncertainty estimates and independently confirmed targets would be needed to shorten this interval, address the limitations noted in both demonstrations and make the results quantitative.

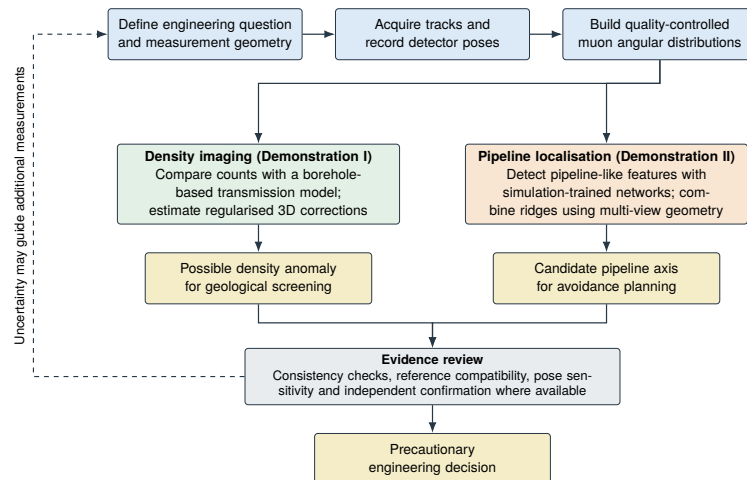


Figure 6. Proposed field-to-decision workflow for tunnel muography.

6 CONCLUSIONS

A three-layer Micromegas muography system operated on two TBMs of Shenzhen Metro Line 22 over 1371 rings. It ran unattended for months with 97% availability once external outages were excluded, and although TBM advance increased the raw data rate, three-layer track reconstruction kept the muon measurement unaffected. On the TBM, muon data from two rings extended the borehole-derived density model into three dimensions and indicated a possible low-density zone above the tunnel. From a completed parallel tunnel, a four-station survey estimated the axis of a buried pipeline that conventional methods had failed to locate; the tunnel alignment was modified as a precaution and the shield passed the crossing without interference, although the pipeline position was not verified directly. Because each measurement currently needs about one day of exposure, muography is suited to targeted screening of known concerns rather than continuous warning. Shorter exposures, matched sky references, systematic pose records and reconstruction with uncertainty estimates are the next steps towards routine use in shield tunnelling.

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